NATIONAL RESEARCH UNIVERSITY
HIGHER SCHOOL OF ECONOMICS

Faculty of Computer Science
Bachelor's Programme "Data Science and Business Analytics"

Research Project Report on the Topic:
A/B Testing of a Recommender System as a Black-Box Experiment

Submitted by the Student:

group #БПАД244, 2nd year of study — Sapunov Artemiy Maximovich

Approved by the Project Supervisor:

Ramazyan Tigran Armenovich
Research Fellow
Faculty of Computer Science, HSE University

Moscow 2026

Contents

Abstract

This project looks at how a change in a recommender system feature affects user behavior, using a standard A/B test run in an online setting. The joint system "recommender + user" is treated as a black box, and the analysis focuses exclusively on the observable input–output relationship: the input is the assignment of a user to the control version A or the treatment version B, while the outputs are product metrics (e.g., CTR, conversion, retention) computed from logged user events. The planned outcome of the work is a formalized description of the experiment, including the problem statement, the experimental methodology (unit of randomization, recommendation display policy, and experiment data model), a pre-determined hierarchy of metrics (primary, secondary, and guardrail metrics), and a statistical analysis plan for estimating the treatment effect and making a decision about implementation. The project is positioned with respect to classical and practical literature on reliable online experimentation, addressing common problems such as sample ratio mismatch (SRM) and the interpretation of multiple metrics.

Аннотация

Проект посвящён оценке изменения рекомендательной системы через онлайн рандомизированный эксперимент (A/B-тест). Система рассматривается как «чёрный ящик»: входом является назначение пользователю варианта A (control) или B (treatment), а выходом — продуктовые метрики, рассчитанные по логам событий (CTR, конверсия, удержание и др.). Планируемый результат работы — формализированное описание эксперимента: постановка задачи, методология проведения эксперимента (единица рандомизации, политика показа рекомендаций, модель данных эксперимента), заранее фиксируемая иерархия метрик (primary/secondary/guardrails) и план статистического анализа для оценки эффекта и принятия решения о внедрении. Работа опирается на практические источники по надёжным онлайн-экспериментам и типовым проблемам, включая SRM и интерпретацию множества метрик.

Keywords

A/B testing; online controlled experiments; recommender systems; black-box evaluation; randomization; CTR; conversion; retention; SRM; guardrail metrics

1 Introduction

1.1 Subject Area and Motivation

Modern digital products continuously improve user experience through stepwise changes to ranking, UI blocks, and recommendation logic. Offline evaluation of recommenders (e.g., ranking metrics on previously collected data) may not reflect real user behavior and business outcomes; therefore, online randomized experiments remain the most reliable decision instrument for product releases.

1.2 Problem Statement

We consider an existing product that displays recommendation blocks to users. A new version of the recommender feature (version B) is proposed. The goal is to determine whether version B improves a pre-defined product objective compared to the current version A.

Statement. Run an A/B test where users are randomly assigned to A or B. Using event logs, compute agreed metrics and decide whether to ship B.

1.3 Planned Outcomes

By the end of the project, the report will contain:

- a formalized description of the experiment, including the experimental methodology (unit of randomization, recommendation display policy, and experiment data model);
- a pre-determined hierarchy of metrics, including primary, secondary, and guardrail metrics, and SRM checks;
- a statistical analysis plan for estimating the treatment effect, including estimator choice, confidence intervals or hypothesis tests, handling of multiple metrics, and the decision rule;
- a decision framework for implementation (approve deployment, reject the change, or schedule additional iterations), with risks and limitations.

2 Literature Review

This project relies on established work on online A/B testing and causal inference. The reviewed literature explains why randomized experiments can be used to measure the causal effect of product changes and how such experiments should be designed and analyzed in practice.

Rubin (1999) introduces the potential outcomes framework and shows that randomized assignment is sufficient for causal interpretation [6].

Kohavi and Longbotham (2016) describe how A/B tests are conducted in real online systems, including variant definition, randomization, and common implementation issues [5].

Deng et al. (2017) emphasize that statistical analysis must align with the unit of randomization to avoid underestimating variance and overstating significance [1].

Johari et al. (2017, 2022) show that continuous monitoring can invalidate classical tests and propose always-valid inference tools [4, 3].

Fabijan et al. (2019) analyze sample ratio mismatch (SRM) and motivate diagnostic checks as a prerequisite to outcome analysis [2].

Overall, these studies show that A/B testing can provide reliable evidence for product decisions when experiments are carefully designed, analyzed, and monitored.

3 Plan of Further Work

3.1 Step 1: Experimental Object and Context

We fix what is being tested: where recommendations appear, how impressions and clicks are defined, and how versions A and B differ in observable user experience. The unit of analysis is chosen and justified.

3.2 Step 2: Metric System and Pre-registration

We define in advance how success is measured: one primary metric, secondary metrics for interpretation, and guardrails to prevent regressions. SRM diagnostics and actions are specified beforehand.

3.3 Step 3: Data and Logging

We describe the required event data and logging fields (assignment, timestamps, actions), and define basic data quality checks.

3.4 Step 4: Experimental Design

We specify randomization, traffic split, and a fixed-horizon stopping rule. Potential threats (seasonality, novelty) are discussed.

3.5 Step 5: Statistical Analysis

We pre-define the estimator, uncertainty quantification, and decision rule. Secondary and segment analyses are treated as exploratory.

3.6 Step 6: Final Report and Outcomes

We combine design, analysis, results, and limitations, and end with a decision and next steps.

Implementation Overview

The practical component of this project is a Python package `ab_blackbox`, consisting of five modules and three executable scripts. Table 1 summarises all ten files and their roles.

Table 1 — Project files and their roles

File	Layer	Role
`ab_blackbox/model.py`	Package	Symbolic response models (SymPy)
`ab_blackbox/simulator.py`	Package	Black-box oracle
`ab_blackbox/experiment.py`	Package	A/B experiment runner
`ab_blackbox/analysis.py`	Package	Statistical analysis pipeline
`ab_blackbox/datasets.py`	Package	Data loading and calibration
`ab_blackbox/__init__.py`	Package	Public API
`demo.py`	Script	Synthetic end-to-end demonstration
`main.py`	Script	Real-data pipeline
`optimize_demo.py`	Script	Black-box optimisation on synthetic data
`optimize_main.py`	Script	Black-box optimisation calibrated from real data

4 Experimental Design

The experiment follows a standard two-variant randomised design. Users are assigned independently and uniformly at random to one of two variants: control (A) or treatment (B). Assignment is performed at the user level to ensure that each user receives a consistent experience throughout the experiment.

The *experimental object* is a button displayed to users on a webpage. The treatment modifies button parameters—colour, size, and label text—each encoded as a continuous value in $[0, 1]$, where 0 represents the current default and 1 represents the fully activated alternative. The control variant corresponds to $(\texttt{color}, \texttt{size}, \texttt{text}) = (0, 0, 0)$; the treatment in the baseline experiment sets $\texttt{color} = 1$ while holding the remaining parameters at zero, matching the real-world dataset structure.

The joint system of recommender (or UI feature) and user is treated as a *black box*: the only observable quantities are the input variant assignment and the output event log. The internal mechanism is not inspected. A single call to the black box simulates the full experiment for a given parameter configuration and returns aggregate metrics.

The experiment duration and sample size are fixed in advance. The dataset used for calibration contains $n = 60{,}000$ users per group, which substantially exceeds the analytically derived minimum of $n = 9{,}458$ per group required to detect the observed effect at 80% power with $\alpha = 0.05$ (Section 7).

5 Metric System

Three tiers of metrics are defined prior to data collection, following the pre-registration principle.

Primary metric. Click-through rate (CTR), defined as the fraction of users who clicked the button at least once:

$$\widehat{\mathrm{CTR}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}[\mathrm{clicks}_i \geq 1].$$

CTR is the metric most directly affected by button design and is the basis for the hypothesis test and the decision rule.

Secondary metric. Conversion rate, defined as the fraction of users who both clicked and completed the target action (purchase or sign-up). It is used to interpret the direction and magnitude of the CTR effect. Secondary metric results are treated as exploratory and are subject to Bonferroni correction when reported alongside the primary metric.

Guardrail metrics. The sample ratio $\hat{r} = n_A/(n_A + n_B)$ is monitored as a data-quality guardrail. A significant deviation from the expected split of 0.5 (detected by a chi-square test at $\alpha = 0.01$) indicates a fault in the assignment or logging pipeline and invalidates all downstream analysis.

6 Data Quality and Logging

The dataset `ab_test_results_aggregated_views_clicks_2.csv` contains 120,000 rows with four columns: `user_id`, `group`, `views`, and `clicks`. Groups are labelled `control` and `test`. The dataset contains no missing values and is perfectly balanced ($n_A = n_B = 60{,}000$), so no SRM is present.

Data quality checks implemented in `analysis.py` include:

- SRM chi-square test (run before any metric analysis);
- verification that `Variant` takes only the values `A` and `B`;
- confirmation that `Clicks` and `Conversions` are non-negative integers.

7 Statistical Analysis Plan

Primary test. The treatment effect on CTR is estimated using a two-proportion z-test. Let p_A and p_B denote the true click probabilities under control and treatment. The null hypothesis is $H_0 : p_B = p_A$ against the two-sided alternative $H_1 : p_B \neq p_A$. The test statistic is

$$z = \frac{\hat{p}_B - \hat{p}_A}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_A} + \frac{1}{n_B}\right)}}, \qquad \hat{p} = \frac{x_A + x_B}{n_A + n_B},$$

where x_A, x_B are the observed click counts. The 95% confidence interval on $\Delta = p_B - p_A$ uses the unpooled standard error:

$$\Delta \pm z_{0.025}\sqrt{\frac{\hat{p}_A(1-\hat{p}_A)}{n_A} + \frac{\hat{p}_B(1-\hat{p}_B)}{n_B}}.$$

Minimum sample size. The required per-group sample size for a two-proportion z-test at significance level α and power $1-\beta$ is

$$n = \frac{(z_{\alpha/2} + z_\beta)^2\big(p_A(1-p_A) + p_B(1-p_B)\big)}{\delta^2},$$

where $\delta = |p_B - p_A|$. This formula is derived symbolically in `model.py` using SymPy and evaluated numerically at the calibrated parameter values.

Bootstrap robustness check. A non-parametric bootstrap confidence interval (2000 resamples) is computed alongside the z-test as a robustness check. Agreement between the two intervals confirms that the z-test assumptions are satisfied.

Multiple testing. When the primary and secondary metrics are both reported, Bonferroni correction is applied, adjusting the significance threshold to $\alpha/2 = 0.025$.

Stopping rule. The experiment runs for a fixed pre-determined duration. No interim analysis is performed, which avoids the inflated type-I error that arises from continuous monitoring [4].

8 Risks and Limitations

Novelty effect. Users may interact with the new button variant simply because it is unfamiliar. This inflates the short-term CTR estimate for the treatment and may not persist. The experiment duration should be long enough for novelty to decay.

Assumed parameter weights. The real dataset contains only two conditions (control and test), corresponding to `color` $= 0$ and `color` $= 1$. The weights for `size` and `text` are not observed and are set equal to the observed colour effect ($w = 0.0147$) as a conservative assumption. Any conclusion about optimal parameter combinations other than colour is therefore model-dependent.

Binary click proxy. The dataset records only whether a user clicked, not whether they converted downstream. The conversion metric is approximated as identical to CTR, which underestimates the true conversion funnel.

Synthetic simulation. The black-box simulator generates user behaviour from a parametric model. Real user behaviour may exhibit heterogeneity, autocorrelation, and non-linearity not captured by the linear or logistic response surfaces.

9 Decision Rules

The decision framework has five outcomes, determined sequentially:

1. **INVALID**: SRM detected ($p < 0.01$). Halt analysis; investigate assignment and logging.
2. **SHIP**: Primary metric significantly positive and secondary metric not significantly negative. Deploy variant B.
3. **REJECT**: Primary metric significantly negative. Do not deploy; investigate.
4. **REVIEW**: Primary positive but secondary regressed. Requires manual review of trade-off.
5. **INCONCLUSIVE**: Primary not significant. Increase sample size or experiment duration.

10 Assumptions

The analysis rests on four assumptions inherited from the potential outcomes framework [6]:

1. **SUTVA**: The potential outcome for each user depends only on that user's own assignment, not on other users' assignments (no interference).

2. **Ignorability**: Assignment is independent of potential outcomes given observed covariates. Satisfied by construction under random assignment.

3. **Positivity**: Each user has a non-zero probability of being assigned to either variant. Satisfied by the 50/50 split.

4. **Stable response surface**: The symbolic model in `model.py` assumes that click probability is a deterministic function of button parameters plus i.i.d. noise. Interactions between parameters are not modelled.

Experimental Results

The analysis was run on the calibrated simulator (grounded in real data) via `main.py`. Table 2 reports the key metrics for both the direct test on real data and the calibrated simulation.

Table 2 — A/B test results: real data and calibrated simulation

Source	**CTR A**	**CTR B**	**Δ**	***p*-value**
Real data (direct z-test)	0.1470	0.1617	+0.0147	< 0.0001
Calibrated simulation ($n = 60{,}000$)	0.1476	0.1682	+0.0206	< 0.0001

For the direct test on real data: $z = 7.055$, 95% CI on Δ: [+0.0106, +0.0188], relative lift +10.0%. The bootstrap confidence interval [+0.0165, +0.0246] from the simulation excludes zero and agrees directionally with the z-test. No SRM was detected ($p = 1.000$ for the chi-square test on the 60,000 / 60,000 split). The decision rule yields **SHIP**.

Required sample size. The minimum per-group sample size to detect the observed effect ($\delta = 0.0147$) at 80% power and $\alpha = 0.05$ is $n = 9{,}458$, derived analytically from the SymPy model. The actual dataset ($n = 60{,}000$) is approximately 6× larger than required, making the experiment highly powered.

Black-box optimisation. Running `optimize_main.py` sweeps over all combinations of (`color`, `size` $\{0, 0.25, 0.5, 0.75, 1.0\}^3$, totalling 125 black-box calls. The optimal configuration found by grid search is (`color` $= 1$, `size` $= 1$, `text` $= 1$) with a simulated CTR of 0.1944, representing a +20.2% additional lift over the real experiment's treatment CTR of 0.1617. Table 3 compares the three search strategies.

Table 3 — Optimisation results: CTR and black-box call budget

Method	**Calls**	**Best CTR**	**Lift vs. real test**
Real experiment (A→B)	—	0.1617	(reference)
Grid search	125	0.1944	+20.2%
Random search (30 calls)	30	0.1860	+15.0%
Greedy 1-D sweep	15	0.2000	+23.7%
SymPy theoretical	0	0.1912	+18.2%

Discussion: Interpretation and Effects

Treatment effect. The test variant produces a statistically significant increase in CTR of $+0.0147$ (relative lift $+10.0\%$) over the control. The effect is robust: it is confirmed by both the parametric z-test and the non-parametric bootstrap, and the 95% confidence interval is entirely positive. With $n = 60{,}000$ per group the experiment is well-powered and the estimate is precise.

Black-box perspective. Treating the recommender–user system as a black box is methodologically appropriate: we make no assumptions about the internal ranking algorithm, user psychology, or platform architecture. The causal interpretation relies only on random assignment [6], which is satisfied by design.

Optimisation vs. single test. The standard A/B test compares two fixed configurations. The optimisation component extends this to a search over the parameter space. The SymPy symbolic model provides the theoretical optimum at zero query cost, while the simulator allows empirical validation under sampling noise. The comparison in Table 3 shows that random search with 30 calls recovers 74% of the additional lift achieved by exhaustive grid search with 125 calls, illustrating the efficiency gain from sub-exhaustive methods.

Limitations. The weights for `size` and `text` are not observed from data and are assumed equal to the colour weight. Conclusions about the optimal multi-parameter configuration are therefore conditional on this assumption. A follow-up experiment varying size and text independently would allow the model to be re-calibrated with observed values, removing the assumption.

Novelty and long-run effects. The dataset covers a fixed observation window. If the CTR improvement is driven partly by novelty, the long-run effect may be smaller than estimated. Monitoring post-deployment CTR over several weeks would confirm whether the effect persists.

Conclusion

This project demonstrated a complete black-box A/B testing pipeline, from symbolic response modelling (SymPy) through simulation, statistical analysis, and parameter optimisation. The treatment variant produces a statistically significant +10.0% lift in CTR, the decision rule yields **SHIP**, and the optimisation component identifies configurations with up to +20.2% additional lift using as few as 15–30 black-box calls.

References

[1] Alex Deng, Jiannan Lu, and Jennifer Litz. “Trustworthy Analysis of Online A/B Tests: Pitfalls, Challenges and Solutions.” In: *Proceedings of the Tenth ACM International Conference on Web Search and Data Mining (WSDM)*. 2017.

[2] Aleksander Fabijan, Pavel Dmitriev, Jay Gupchup, Jason Omhover, and Lukas Vermeer. “Diagnosing Sample Ratio Mismatch in Online Controlled Experiments.” In: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2019.

[3] Ramesh Johari, Peter Koomen, and Leonid Pekelis. “Always Valid Inference: Continuous Monitoring of A/B Tests.” In: *Operations Research* (2022).

[4] Ramesh Johari, Peter Koomen, Leonid Pekelis, and David Walsh. “Peeking at A/B Tests: Why It Matters, and What to Do About It.” In: *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2017.

[5] Ron Kohavi and Roger Longbotham. “Online Controlled Experiments and A/B Testing.” In: *Encyclopedia of Machine Learning and Data Mining*. Springer, 2016.

[6] Donald B. Rubin. “Teaching Causal Inference in Experiments and Observational Studies.” In: *Proceedings of the American Statistical Association, Section on Statistics Education* (1999).